Michael C.

Senior Java Backend Engineer — Payments, Wallet and Ledger Systems

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SUMMARY

- 13 years of Java backend at large-scale companies (Alibaba, Didi); the last 4 years building systems that move real money: wallets, deposits, withdrawals and reconciliation at Huobi and u8.com.
- Strong on what iGaming platforms get hurt by: concurrent balance deduction, duplicate requests, lost payment callbacks, payout safety and daily reconciliation.
- Owns a module from design to production. Written-first, async-friendly, works well remotely.

EXPERIENCE

u8.com — Backend Owner, Player Wallet and Payments **2024.04 – Present**

Online iGaming platform.

Owned the wallet core path: **deposit credit → withdrawal payout → daily reconciliation**. Every balance change goes through the wallet service ledger.

- **Concurrency-safe balances (no overdraw):** concurrent deductions queue on a row lock; each is a single conditional UPDATE that checks and deducts the balance, so it never goes negative.
- **Idempotent payment API:** one-time Redis token on the payment page, with a unique business order number as backstop. Double clicks and retries are booked once.
- **Lost-callback recovery:** verify signature and amount before booking a channel notification; a scheduled job queries the channel to fill in missed callbacks.
- **Safe withdrawals:** freeze the balance, then ask the channel to pay out. On timeout, query only and never resend, so a player is never paid twice. Deduct on confirmed success, unfreeze on confirmed failure.
- **Eventual consistency:** the balance change and its outbound notification are written in one database transaction (local message table), then delivered via Kafka with retry and idempotent consumers.
- **Daily reconciliation:** match channel statements to the internal ledger order by order; auto-book missing records, route extras to manual review.

Huobi Technology — Backend Owner, Blockchain Wallet **2022.05 – 2024.04**

Major global cryptocurrency exchange; the wallet is a decentralized multi-chain wallet with about 12M users.

- Owned wallet creation, on-chain transfers, deposit detection and push notifications.
- Block scanning with chain-fork handling (parentHash check, confirmation count) and high availability (Redis distributed lock, resume from a saved cursor).
- Reliable notification delivery and idempotent consumption on Kafka: one notification per deposit.
- Fixed a recurring scanner OOM by replacing an unbounded queue with a bounded one; Bloom filter to cut lookups across thousands of addresses per block.

Didi Chuxing — Senior Java Backend Engineer, Search **2018.10 – 2022.05**

China's leading ride-hailing platform (often called the Uber of China).

- Tech lead. Built the company-wide search service on Elasticsearch from scratch (weekly requests 60K → 100K) and a SQL-dependency query service used across the company.

Alibaba Group — Java Backend Engineer, Distributed Scheduling (DataWorks) **2016.10 – 2018.10**

China's largest e-commerce and cloud company (NYSE-listed); DataWorks is Alibaba Cloud's data platform.

- Core development on the distributed scheduler: master election, task sharding, Quartz timers running 1M+ tasks. Average task development time fell from 36 to 17.45 hours.

Zhangqu Technology — Java Backend Engineer, Mobile Game Operations Backend 2013.03 – 2016.10

Mobile game company listed on China's ChiNext (A-share) market.

- Built the game log-collection SDK shared by several mobile titles, plus the Java operations back office and scheduled jobs.

SKILLS

Java, Spring / Spring Boot, MySQL, Redis, Kafka · Payments: idempotency, state machines, channel integration, reconciliation · Blockchain wallets · Elasticsearch

EDUCATION

M.S. Artificial Intelligence — Hawaii Pacific University (USA) 2025.01 – 2026.07

B.S. Computer Science and Technology — University of Electronic Science and Technology of China 2008.09 – 2012.07